

# Fixed and Moving Local References in the Rotemberg–HSA Phillips Curve

Restriction Version: F0, U, R1, R2, and R3

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Automatically built from provenance-validated production runs

## Abstract

This restriction-version report is separate from `nkpc_hsa_report.pdf`, which remains the historical reduced-form and ablation report. The present document asks one question: does the local Rotemberg–HSA cross-equation restriction survive contact with the data? It compares a fixed-reference local benchmark (F0) with an unrestricted moving-reference model (U), the cross-restricted model (R1), its weak Third-law extension (R2), and a constant-pass-through benchmark (R3), on 124 quarters from 1982Q1 to 2012Q4. Every numerical table, figure and in-text number is generated from the signed theory-run namespace; historical artifacts are never substituted for a missing current run.

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# 1 Why we estimate this equation: the local Rotemberg–HSA Phillips curve

## 1.1 The HSA Phillips curve

The mechanism under test is that the slope of the Phillips curve depends on how many firms compete. In a Rotemberg economy with a CES demand structure, the slope of the inflation equation with respect to real marginal cost is governed by the demand elasticity  $\zeta$  and the adjustment-cost parameter  $\chi$ , through  $\kappa = (\zeta - 1)/\chi$ . The HSA extension makes  $\zeta$  itself a function of the number of competitors  $N$ : entry raises the elasticity each firm faces, so a market with more firms passes cost shocks into prices faster. Falling firm counts then flatten the Phillips curve, which is the empirical regularity this project was built to assess.

Three statements are logically distinct, and this report keeps them apart throughout. The *Second law* says that competition raises pass-through:  $\theta_0 > 0$  and  $\kappa_N > 0$ . The *weak Third law* says that the entry channel itself weakens as the reference level of competition rises:  $\gamma \leq 0$ . The *cross-equation restriction* says something stronger than either — that the same structural object governs both channels, so the slope response and the entry coefficient are not free to move independently.

## 1.2 Two ways competition enters

The estimating equation lets competition act through two separate objects, and the distinction is what makes a cross-equation restriction meaningful at all.

The *reference level*  $\bar{N}_t$  is a slow-moving local anchor: the level of competition around which the firm’s problem is linearised. It shifts the coefficients themselves,  $\kappa_t = \kappa_0 + \kappa_N \bar{N}_t$  and  $\theta_t = \theta_0 + \gamma \bar{N}_t$ . The *cycle*  $\hat{N}_t$  is the deviation of observed competition from that anchor, and it enters the inflation equation directly as a regressor with coefficient  $\theta_t$ .

F0 shuts the first channel off. It fixes the reference at a single  $N_0$ , so  $\kappa_t = \kappa_0$  for every period and only the cycle matters. Everything the restricted models say about  $\kappa_N$  is therefore invisible in F0, which is why F0 is a benchmark and not a nested special case.

## 1.3 From theory to the estimating equation

With  $\bar{N}$  measured in ten log points from the anchor ( $\bar{N} = 10(\log N - \log N_0)$ , anchor  $N_0 = 7.862$  in the raw units of the competition series) and inflation in percentage points, the quarterly structural identity  $d\kappa/d\log N = \zeta\Theta$  maps into the stored empirical coefficients as

$$\frac{d\kappa}{d\log N} = 10\kappa_N, \quad 100\kappa_N = b_x \zeta_0 \theta_0.$$

The second equality is the cross-equation restriction. It is imposed with  $\zeta_0 = 6.00$  held fixed (implying a flexible-price markup  $\mu_0 = 1.20$ ) and an explicitly configured marginal-cost loading  $b_x = 1.00$ . The loading is never defaulted to one: a regression on an activity *gap* would need its own  $b_x$ , and the code refuses to impose the identity without one.

## 1.4 Distance from the theory

The moving reference is where this report departs from Eq. 21 of the underlying theory. The theory is a fixed-reference local approximation; letting  $\bar{N}_t$  drift is a semi-structural device that keeps the local approximation near the data as competition trends over three decades. The

taxonomy below states, for each object in the model, whether it is a theory identity, a maintained law, a domain condition, an empirical approximation, or something deliberately left free.

| Category                                  | Contents                                                                                                                                                         |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theory identity / cross-equation relation | $\kappa_0 = (\zeta_0 - 1) / \chi$ ; $\theta_0 = (1 / \chi) * (1 - \rho_0) / \rho_0$ ; $\mu_0 = \zeta_0 / (\zeta_0 - 1)$ ; $d \kappa / d \log N = \zeta * \theta$ |
| Maintained HSA law                        | Second law: $\theta_0 > 0$ and $\kappa_N$ empirical $> 0$ ; Weak Third law: $\gamma \leq 0$ (R2 only)                                                            |
| Domain / local admissibility              | $\chi > 0$ ; $\zeta_0 > 1$ ; $\mu_0 > 1$ ; $\kappa_0 > 0$ ; optional whole-path $\kappa_t > 0$ and $\theta_t > 0$                                                |
| Moving-reference empirical approximation  | moving $N_{star\_t}$ is a slow-moving local reference; $\kappa_t = \kappa_0 + \kappa_N$ empirical $* N_{bar\_t}$ ; $\theta_t = \theta_0 + \gamma * N_{bar\_t}$   |
| Deliberately not imposed (future R4)      | $\kappa_0 = (\zeta_0 - 1) / \chi$ ; $\theta_0 = (10 / \chi) * (1 - \rho_0) / \rho_0$                                                                             |

Two identities of the fixed-reference theory —  $\kappa_0 = (\zeta_0 - 1)/\chi$  and  $\theta_0 = (10/\chi)(1 - \rho_0)/\rho_0$  — are deliberately *not* imposed. Imposing them would pin  $\chi$  from the level of the slope, and the level of the slope is exactly the quantity the moving-reference approximation is least entitled to. A model that imposes them is future work (R4), not a variant estimated here.

## 2 Data

Every run in this report uses one data cell, so that a difference between two rows of any table is a difference in restrictions and never in data. The cell pairs quarterly CPI inflation with a Beveridge–Nelson inverse-markup gap as the real marginal cost proxy, and an annual firm count observed in Q4.

| Series                            | Column          | Transformation                                | Mean    | SD     | Min     | Max    |
|-----------------------------------|-----------------|-----------------------------------------------|---------|--------|---------|--------|
| Inflation $\pi_t$                 | pi_cpi_qoq      | exact_qoq_percent_change_<br>nonannualized    | 0.7323  | 0.4979 | -2.2902 | 1.7367 |
| Lagged inflation $\pi_{t-1}$      | pi_cpi_qoq_prev | one-quarter lag of the same<br>transformation | 0.7400  | 0.5042 | -2.2902 | 1.7367 |
| Expected inflation $E_t\pi_{t+1}$ | Epi_qoq         | annual_percent_divided_by_4                   | 0.7771  | 0.2886 | 0.0312  | 1.6012 |
| Real marginal cost $x_t$          | markup_BN_inv   | BN_inverse_markup_gap                         | -0.0656 | 0.6478 | -1.4584 | 1.7483 |
| Competition proxy $N_t$           | N_Gustavo       | log100_centered10, observed<br>annual_q4      | 7.9556  | 1.2101 | 5.9982  | 9.6786 |

Two features of the competition series matter for what follows. It is observed annually while the model is quarterly, so 124 quarters carry roughly one competition observation in four; the state-space treatment below handles the missing quarters as missing, not as interpolated data. And the series trends: the moving reference in Figure 5 travels several ten-log-point units over the sample, which is precisely the range over which a *local* linear approximation has to be defended rather than assumed.

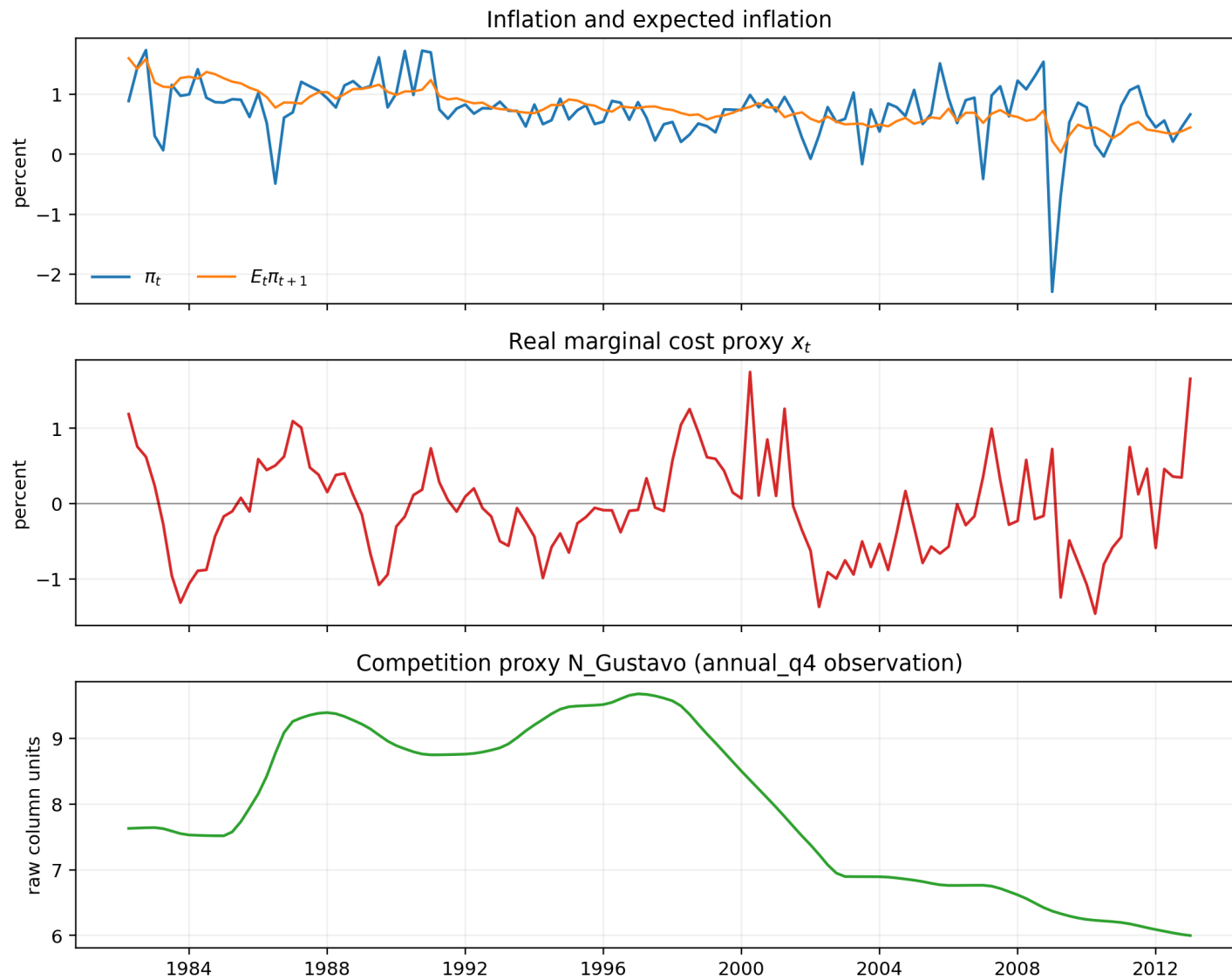


Figure 1: The observed series behind every restriction run.

### 3 Econometric model and estimation

#### 3.1 Model variants

Five models are estimated. Within the moving-reference family the nesting is  $U \supset R1 \supset R2$ , and R3 is  $R1 + \{\gamma = 0\}$ . F0 is not nested in that family, because its reference is fixed rather than restricted.

| Model | Slug   | Reference        | Cross restriction | Second law     | Third law | Gamma          | State sampler        |
|-------|--------|------------------|-------------------|----------------|-----------|----------------|----------------------|
| F0    | hsa_f0 | fixed reference  | no                | maintained     | neither   | not_applicable | fixed reference ffbs |
| U     | hsa_u  | moving reference | no                | not maintained | neither   | unrestricted   | particle gibbs       |
| R1    | hsa_r1 | moving reference | yes               | maintained     | neither   | unrestricted   | particle gibbs       |
| R2    | hsa_r2 | moving reference | yes               | maintained     | weak      | gamma <= 0     | particle gibbs       |
| R3    | hsa_r3 | moving reference | yes               | maintained     | neither   | gamma = 0      | joint ffbs           |

#### 3.2 The inflation equation actually estimated

Write  $y_t = \pi_t - E_t \pi_{t+1}$  and  $a_t = \pi_{t-1} - E_t \pi_{t+1}$ . The equation carried to the data is

$$y_t = \alpha a_t + \underbrace{(\kappa_0 + \kappa_N \bar{N}_t)}_{\kappa_t} x_t - \underbrace{(\theta_0 + \gamma \bar{N}_t)}_{\theta_t} \hat{N}_t + \lambda \zeta_t + \eta_t, \quad \zeta_t = x_t - \phi_1 x_{t-1},$$

with  $\eta_t \sim N(0, \sigma_\eta^2)$ . The term  $\lambda \zeta_t$  carries the correlation between the inflation disturbance and the activity innovation, so  $\sigma_\eta^2$  is the conditional variance and  $\sigma_e^2 = \lambda^2 \sigma_\zeta^2 + \sigma_\eta^2$  the total.

Under the cross-equation restriction the two competition regressors collapse into one. Substituting  $\kappa_N = b_x \zeta_0 \theta_0 / 100$ ,

$$\kappa_N x_t \bar{N}_t - \theta_0 \hat{N}_t = \theta_0 \left[ \frac{b_x \zeta_0}{100} x_t \bar{N}_t - \hat{N}_t \right],$$

so R1–R3 sample one fewer coefficient than U and  $\kappa_N$  is *derived* rather than drawn. This is the sense in which the restriction is exact rather than a shrinkage prior: it removes a dimension from the coefficient block.

#### 3.3 State-space representation

F0 and the moving-reference family use genuinely different state systems.

F0 carries a single competition deviation  $q_t$  around the fixed anchor, with an AR(2) transition and a measurement equation that is active only in the quarters where the annual series is observed. Its state vector is  $(q_t, q_{t-1})$ ; there is no  $\bar{N}$  state to move.

U, R1, R2 and R3 carry both a random-walk-with-drift reference  $\bar{N}_t$  and an AR(2) cycle  $\hat{N}_t$ , observed jointly through  $N_t^{obs} = \hat{N}_t + \bar{N}_t + \nu_t$  with  $\nu_t \sim N(0, \sigma_N^2)$ . Because the reference multiplies the coefficients, the observation equation is bilinear in the states whenever  $\kappa_N \neq 0$  or  $\gamma \neq 0$  — which is why the state update is not a plain FFBS for those models.



### 3.4 Priors

All five models are estimated under one prior specification, **baseline**. Coefficient priors are stated in physical units and converted internally; variance priors for the  $N$  block are in squared ten-log-point units, matched to the scale of the transformed competition series.

| Parameter           | Role                                                   | Prior              |
|---------------------|--------------------------------------------------------|--------------------|
| $\alpha$            | inflation inertia                                      | $N(0.5, 0.2^2)$    |
| $\kappa_0$          | slope at the reference                                 | $N(0.1, 0.2^2)$    |
| $\theta_0$          | entry coefficient                                      | $N(0, 0.2^2)$      |
| $\kappa_N$          | slope response to $\bar{N}$ (U only; derived in R1–R3) | $N(0, 0.02^2)$     |
| $\gamma$            | entry response to $\bar{N}$                            | $N(0, 0.02^2)$     |
| $\phi_1$            | activity AR(1)                                         | $N(0.7, 0.2^2)$    |
| $\rho_1$            | competition cycle AR(2)                                | $N(0.5, 0.2^2)$    |
| $\rho_2$            | competition cycle AR(2)                                | $N(-0.5, 0.2^2)$   |
| $n$                 | reference drift                                        | $N(0, 0.1^2)$      |
| $\sigma_\eta^2$     | inflation equation                                     | $IG(2, 2)$         |
| $\sigma_\zeta^2$    | activity innovation                                    | $IG(0.001, 0.001)$ |
| $\sigma_u^2$        | competition cycle innovation                           | $IG(2, 0.02)$      |
| $\sigma_\epsilon^2$ | reference innovation                                   | $IG(2, 0.01)$      |
| $\sigma_N^2$        | competition measurement                                | $IG(2, 0.01)$      |

One asymmetry deserves to be stated before any table is read. In U,  $\kappa_N$  carries its own prior. In R1–R3 it has none, because it is a deterministic function of  $\theta_0$ ; its prior is whatever the  $\theta_0$  prior induces through  $\kappa_N = b_x \zeta_0 \theta_0 / 100$ . A difference between U’s and R1’s posterior for  $\kappa_N$  is therefore partly a difference in priors, and Appendix C shows exactly how much.

### 3.5 The coefficient block: a truncated-Gaussian coordinate Gibbs

The maintained laws and the domain conditions are hard constraints, not soft penalties:  $\kappa_0 > 0$ ,  $\theta_0 > 0$ ,  $\gamma \leq 0$  in R2, and — when path positivity is required —  $\kappa_t > 0$  and  $\theta_t > 0$  in *every* period. All of these are linear in the coefficient vector, so the conditional posterior is a Gaussian truncated to a polyhedron  $A\beta \geq \ell$ .

Because  $\kappa_t$  and  $\theta_t$  are affine in  $\bar{N}_t$ , requiring positivity at the smallest and largest observed  $\bar{N}$  is equivalent to requiring it at every date, and the constraint matrix needs only those two rows per path. The block is then updated with a coordinate Gibbs sweep that is invariant for the exact truncated conditional. Independent rejection sampling was abandoned here for a reason visible in the diagnostics: the probe column of Table 4 shows that a majority of unconstrained draws would fall outside the admissible region, so a rejection sampler would stall.

### 3.6 The state update: particle Gibbs and the $\gamma = 0$ joint FFBS

For U, R1 and R2 the bilinear observation equation rules out an exact FFBS, and the latent paths are updated by a conditional bootstrap particle Gibbs sweep. For R3,  $\gamma = 0$  makes the joint state linear-Gaussian again, so R3 uses an exact joint FFBS instead; F0 uses an exact FFBS on its own two-state system.

Whole-path positivity is enforced on the state draw by proposing from the unconstrained conditional and keeping the previous path when the proposal violates it. Since the proposal is the exact conditional, this is an independence Metropolis step whose acceptance ratio is one inside the admissible set — the target is the correctly truncated conditional, not an approximation of it. The rejection share is reported, because a high one is evidence that the local approximation is being asked to span too wide an  $N$  range.

### 3.7 Iterations, saved draws, and convergence criteria

The moving-reference models mix far more slowly than F0, and the production lengths reflect measured pilot behaviour rather than a round number: a 12,000 iteration pilot left latent-state ESS in the tens, and a 120,000 iteration follow-up still left  $\rho_2$  short, so U, R1 and R2 run at 400,000 iterations on four chains. A run enters this report only if its diagnostics pass; the builder refuses runs that are quick, dirty-code, stale, or unconverged.

| Model | State sampler        | Coefficient block  | Free coefs | Iterations | Burn  | Thin | Chains | Stored draws |
|-------|----------------------|--------------------|------------|------------|-------|------|--------|--------------|
| F0    | fixed-reference FFBS | positive Gaussian  | 4          | 12000      | 4000  | 5    | 2      | 3200         |
| U     | particle Gibbs       | truncated Gaussian | 6          | 400000     | 50000 | 50   | 4      | 28000        |
| R1    | particle Gibbs       | truncated Gaussian | 5          | 400000     | 50000 | 50   | 4      | 28000        |
| R2    | particle Gibbs       | truncated Gaussian | 5          | 400000     | 50000 | 50   | 4      | 28000        |
| R3    | joint FFBS           | truncated Gaussian | 4          | 36000      | 8000  | 10   | 2      | 5600         |

## 4 Main results

Table 1 collects the restriction-relevant coefficients. Three things stand out, and the rest of the report is an attempt to take each one seriously.

First, the slope level is estimated tightly and consistently:  $\kappa_0$  is 0.0852 in F0, 0.0919 in U, and essentially unchanged across R1–R3. Nothing in the restriction hierarchy disturbs it.

Second, the unrestricted slope response is near zero and two-sided:  $\kappa_N = 0.0006$  in U with a 95% interval of  $[-0.0291, 0.0297]$ . Under the restriction it is positive and bounded away from zero,  $\kappa_N = 0.0068$  with interval  $[0.0011, 0.0178]$  in R1, because the restriction ties it to  $\theta_0$ , which the entry channel does identify.

Third,  $\gamma$  is barely informed by the data. In U it is  $-0.0033$  with interval  $[-0.0380, 0.0297]$ , and Appendix C shows its posterior sitting almost on top of its prior. The weak Third law in R2 therefore does most of its work through the constraint itself, moving  $\gamma$  to  $-0.0143$  with interval  $[-0.0413, -0.0005]$ .

| Model | Parameter         | Mean    | SD     | 2.5%    | 97.5%   | Convergence |
|-------|-------------------|---------|--------|---------|---------|-------------|
| F0    | kappa_0           | 0.0852  | 0.0608 | 0.0042  | 0.2264  | converged   |
| F0    | theta_0           | 0.0446  | 0.0244 | 0.0044  | 0.0963  | converged   |
| U     | kappa_0           | 0.0919  | 0.0540 | 0.0149  | 0.2211  | converged   |
| U     | kappa_N_empirical | 0.0006  | 0.0146 | -0.0291 | 0.0297  | converged   |
| U     | d_kappa_d_logN    | 0.0059  | 0.1455 | -0.2907 | 0.2970  | converged   |
| U     | theta_0           | 0.1261  | 0.0827 | 0.0203  | 0.3378  | converged   |
| U     | gamma             | -0.0033 | 0.0169 | -0.0380 | 0.0297  | converged   |
| R1    | kappa_0           | 0.0846  | 0.0531 | 0.0151  | 0.2136  | converged   |
| R1    | kappa_N_empirical | 0.0068  | 0.0043 | 0.0011  | 0.0178  | converged   |
| R1    | d_kappa_d_logN    | 0.0678  | 0.0430 | 0.0110  | 0.1783  | converged   |
| R1    | theta_0           | 0.1129  | 0.0716 | 0.0183  | 0.2972  | converged   |
| R1    | gamma             | -0.0033 | 0.0166 | -0.0370 | 0.0291  | converged   |
| R2    | kappa_0           | 0.0844  | 0.0530 | 0.0152  | 0.2135  | converged   |
| R2    | kappa_N_empirical | 0.0067  | 0.0042 | 0.0011  | 0.0171  | converged   |
| R2    | d_kappa_d_logN    | 0.0673  | 0.0419 | 0.0112  | 0.1713  | converged   |
| R2    | theta_0           | 0.1122  | 0.0698 | 0.0187  | 0.2856  | converged   |
| R2    | gamma             | -0.0143 | 0.0111 | -0.0413 | -0.0005 | converged   |
| R3    | kappa_0           | 0.0789  | 0.0547 | 0.0085  | 0.2080  | converged   |
| R3    | kappa_N_empirical | 0.0036  | 0.0031 | 0.0001  | 0.0115  | converged   |
| R3    | d_kappa_d_logN    | 0.0359  | 0.0307 | 0.0015  | 0.1152  | converged   |
| R3    | theta_0           | 0.0599  | 0.0511 | 0.0025  | 0.1921  | converged   |

Table 1: Restriction-relevant posterior coefficients.

Restriction-relevant posterior coefficients (90% intervals)

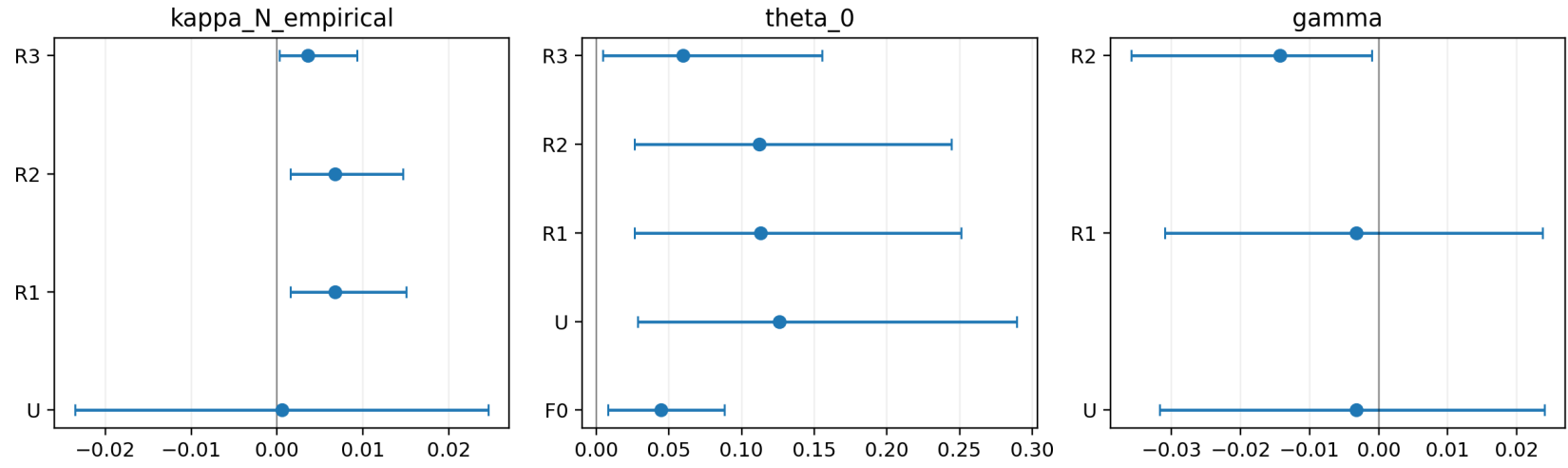


Figure 2: Restriction-relevant posterior coefficients, 90% intervals.

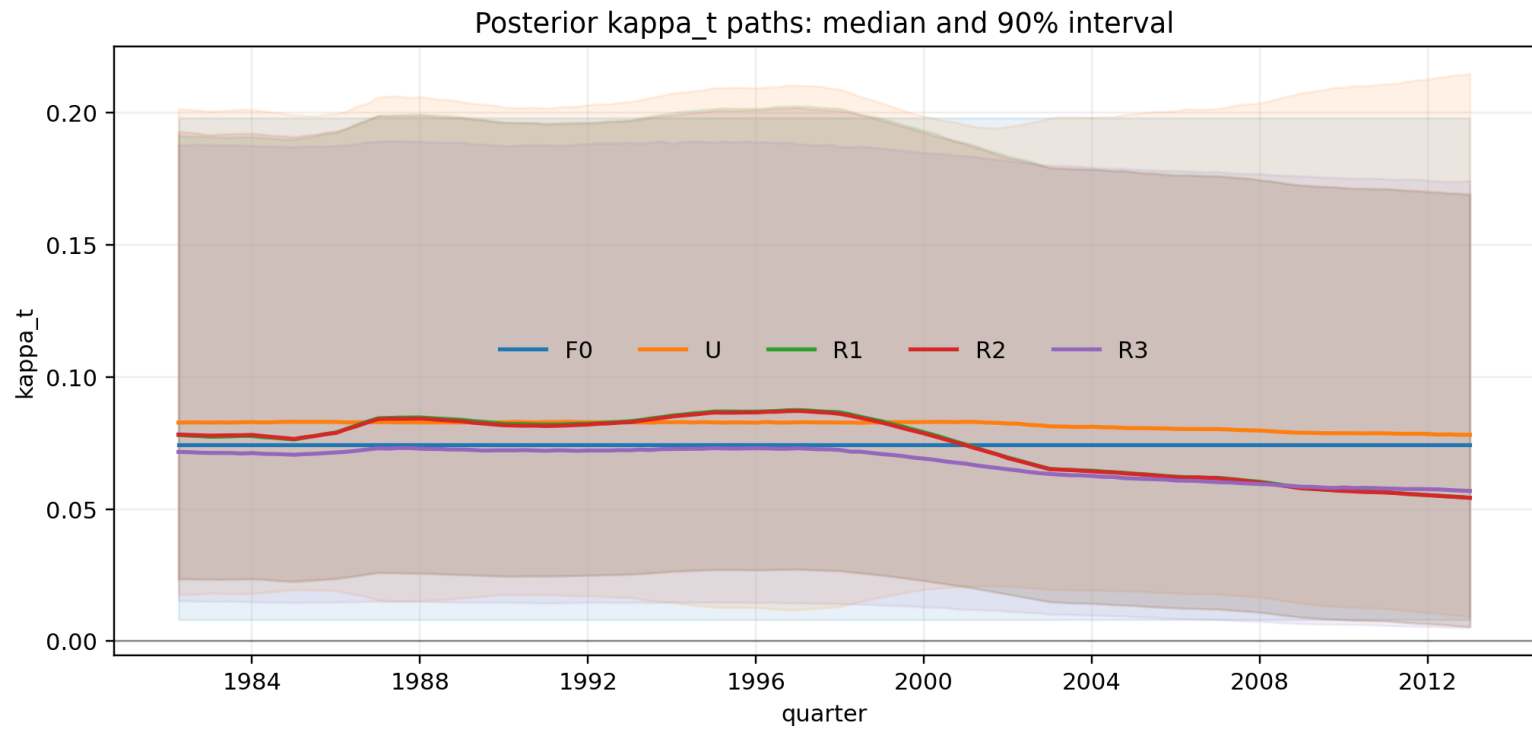


Figure 3: Posterior slope paths  $\kappa_t$  by model.

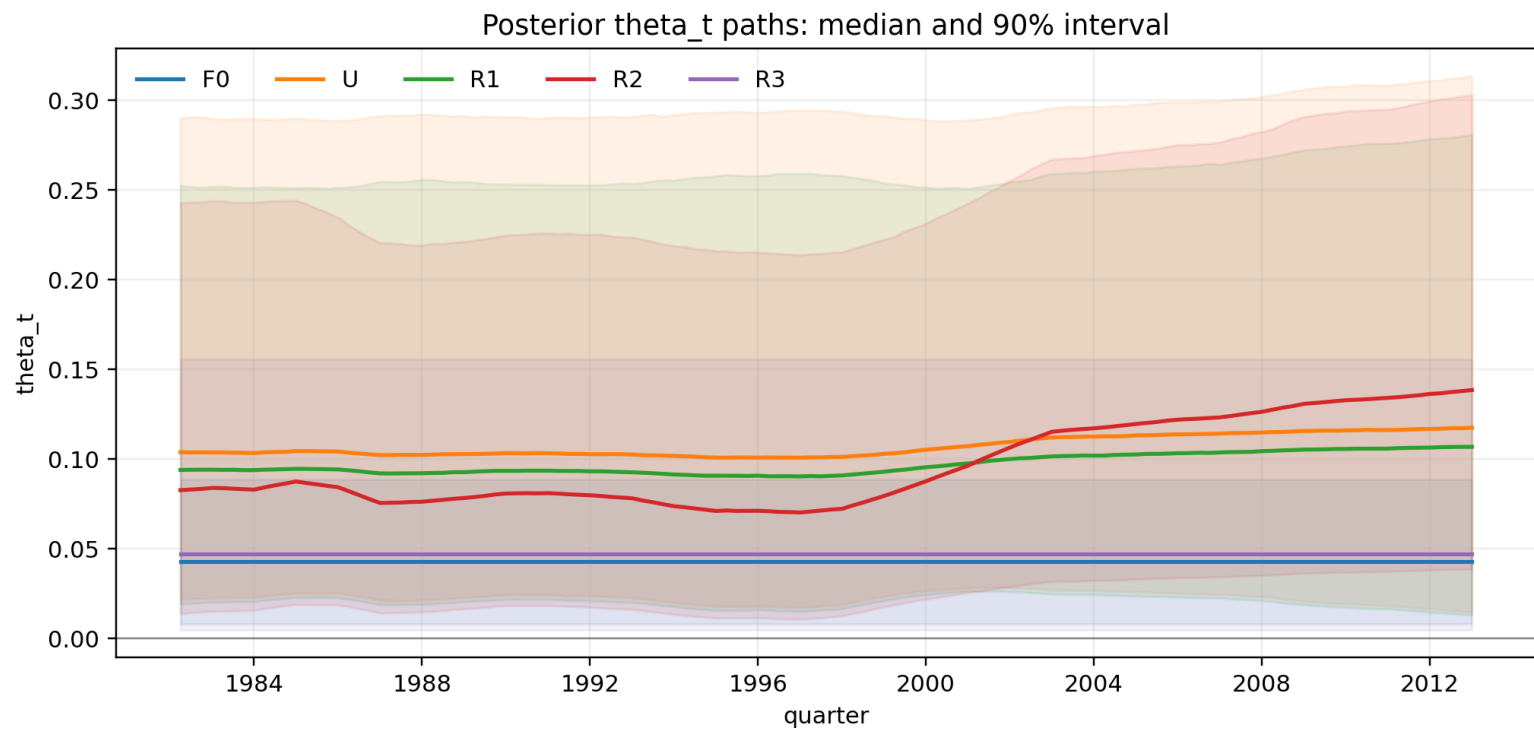


Figure 4: Posterior entry-coefficient paths  $\theta_t$  by model.

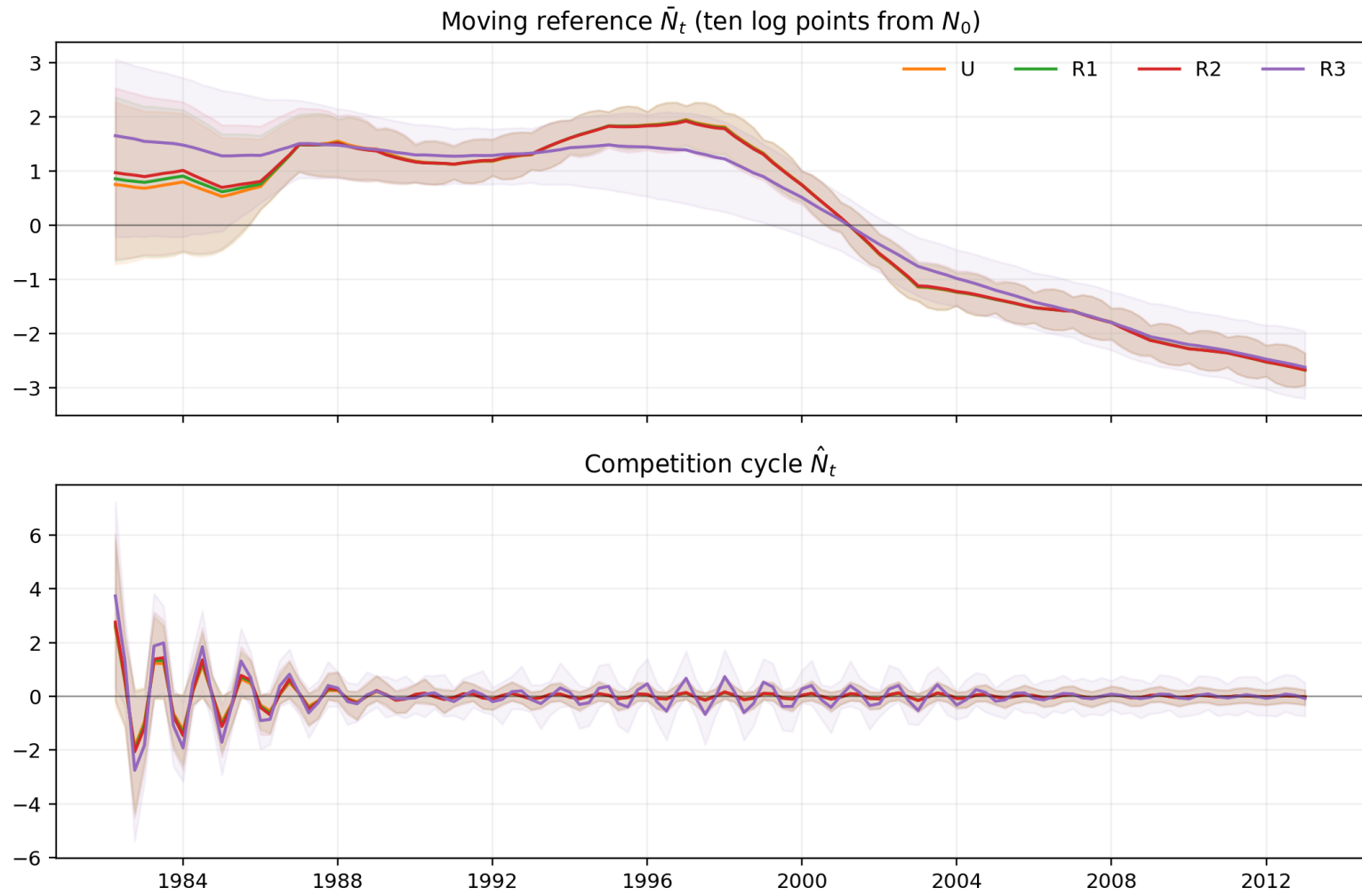


Figure 5: Reference level  $\bar{N}_t$  and cycle  $\hat{N}_t$ , for the models that separate the two.

## 5 The cross-equation restriction against the unrestricted posterior

This is the section the report exists for. U estimates  $\kappa_N$  and  $\theta_0$  separately, so inside U one can ask directly how far the posterior sits from the line the restriction would impose.

At U's own  $\theta_0$ , the restriction would require  $\kappa_N = 0.0076$ . U's posterior mean is 0.0006. The gap has posterior mean  $-0.0070$  with a 90% interval of  $[-0.0327, 0.0181]$  and  $P(\text{gap} > 0) = 0.32$ . The interval contains zero comfortably, so *the data do not reject the restriction*. But the point estimate is an order of magnitude below the restricted value, so neither do the data support it: the unrestricted posterior is simply not informative enough about  $\kappa_N$  to discriminate.

The same conclusion appears more sharply if the identity is inverted. Solving  $100\kappa_N = b_x\zeta_0\theta_0$  for the elasticity gives an implied  $\zeta_0$  with posterior median 0.44 and  $P(\zeta_0 > 1) = 0.48$  — barely better than a coin flip that the implied elasticity is even in the region where a markup exists. The honest reading is that  $\zeta_0$  is not identified by this data cell, which is why the production design fixes it at 6.00 rather than sampling it.

| Model | kappa_N mean | Restriction value | Gap mean | Gap 5%  | Gap 95% | P(gap>0) | Implied zeta median | P(zeta>1) |
|-------|--------------|-------------------|----------|---------|---------|----------|---------------------|-----------|
| U     | 0.0006       | 0.0076            | -0.0070  | -0.0327 | 0.0181  | 0.3181   | 0.4448              | 0.4789    |
| R1    | 0.0068       | 0.0068            | 0.0000   | 0.0000  | 0.0000  | 0.0000   | 6.0000              | 1.0000    |
| R2    | 0.0067       | 0.0067            | 0.0000   | 0.0000  | 0.0000  | 0.0000   | 6.0000              | 1.0000    |
| R3    | 0.0036       | 0.0036            | 0.0000   | 0.0000  | 0.0000  | 0.0000   | 6.0000              | 1.0000    |

Table 2: The cross-equation restriction inside U. R1–R3 satisfy the equality by construction; their rows verify that the constraint is exact, not approximate, and show where it places the pair.



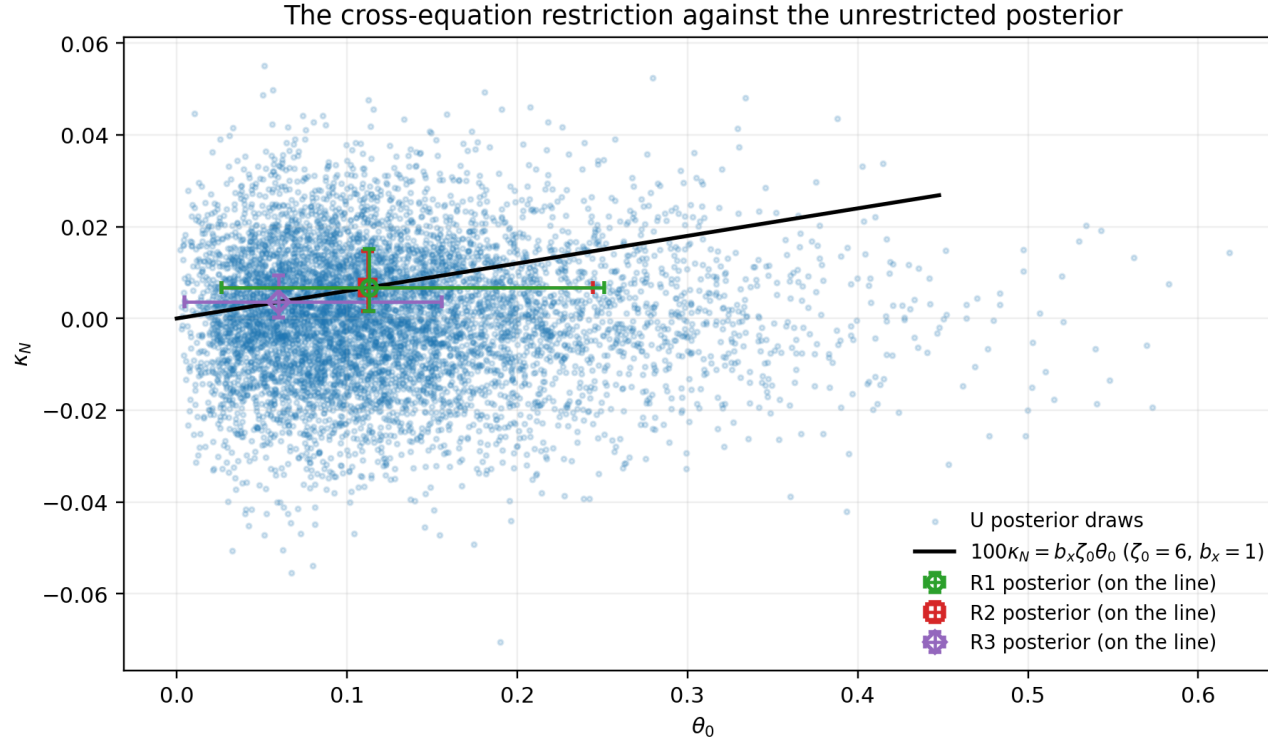


Figure 6: U's  $(\theta_0, \kappa_N)$  posterior cloud against the line  $100\kappa_N = b_x \zeta_0 \theta_0$ , with the restricted posteriors marked on it.

The figure makes the geometry plain. The restriction line passes through the U cloud, but the cloud is wide in the  $\kappa_N$  direction and almost uninformative about its sign, while it is much tighter in  $\theta_0$ . The restriction is not fighting the data; it is supplying identification the data do not have.

## 6 Does the restriction improve fit?

The restriction removes a coefficient. Table 3 asks what that costs in the fit of the inflation equation.

| Model | Free coefs | RMSE   | sigma_eta | Mean log-lik | p_D    | Conditional DIC |
|-------|------------|--------|-----------|--------------|--------|-----------------|
| F0    | 4          | 0.4368 | 0.4685    | -74.2206     | 3.5916 | 152.0327        |
| U     | 6          | 0.4395 | 0.4702    | -74.9305     | 3.9738 | 153.8348        |
| R1    | 5          | 0.4389 | 0.4697    | -74.7577     | 3.7206 | 153.2360        |
| R2    | 5          | 0.4388 | 0.4696    | -74.7286     | 3.6791 | 153.1362        |
| R3    | 4          | 0.4398 | 0.4701    | -75.0018     | 3.7058 | 153.7094        |

Table 3: In-sample fit of the inflation equation, evaluated at the stored latent paths.

**What this table is, and what it is not.** The residuals are computed at each stored draw from the same arithmetic the sampler’s own regression uses, so the numbers are exact for the equation as estimated. But they are *conditional* on the latent competition path: a model with a richer state block is not charged for that richness, and the effective-parameter count  $p_D$  counts coefficient flexibility only. This is a diagnostic of whether imposing the restriction moves the fitted inflation equation. It is not a marginal likelihood, and it must not be read as a Bayes factor. The formal nested comparison is an open item in Section 11.

Read that way, the answer is clean: the restriction is nearly free. Conditional DIC moves from 153.8 in U to 153.2 in R1 — R1 is in fact marginally *better*, because it spends 5 coefficients instead of U’s 6 for the same residual fit. RMSE is 0.439 against 0.439. F0, with the fewest coefficients and no moving reference at all, is not worse than either. On the inflation equation alone, this data cell cannot tell these models apart.

## 7 Q/Q versus four-quarter YoY observation design

The two inflation transformations are alternative observations of one price index, and they are estimated in separate likelihoods — never stacked as if they were independent information.

The restricted models are Q/Q-only in this revision, and this is a identification decision rather than a convenience. The cross-equation identity is a statement about a *quarterly* structural coefficient. A direct four-quarter regression estimates a reduced-form coefficient whose relation to the quarterly one depends on the dynamics being aggregated, so imposing  $100\kappa_N = b_x\zeta_0\theta_0$  on it would attach the structural meaning to the wrong object. The code enforces this: a restricted model with `inflation_observation=yoy_4q` raises rather than runs. F0 and U carry the observation-design comparison instead, because neither imposes the identity.

| Model | Inflation | T   | Status    | kappa0 | theta0 | kappaN  |
|-------|-----------|-----|-----------|--------|--------|---------|
| F0    | Q/Q       | 124 | converged | 0.0852 | 0.0446 | –       |
| U     | Q/Q       | 124 | converged | 0.0919 | 0.1261 | 0.0006  |
| F0    | 4Q YoY    | 124 | converged | 0.1169 | 0.0747 | –       |
| U     | 4Q YoY    | 124 | converged | 0.1159 | 0.1283 | -0.0016 |

The comparison is reassuring for the level of the slope and unhelpful for its response:  $\kappa_0$  and  $\theta_0$  are similar across the two designs, while  $\kappa_N$  changes sign between them in U. That is the same non-identification seen in Section 5, viewed from a second angle.

## 8 Scope condition and the limits of identification

Three limits bound what this report can claim, and each is measured rather than asserted.

**The local approximation spans a wide  $N$  range.** The reference in Figure 5 moves several ten-log-point units over the sample. A local linearisation is being asked to hold across that whole span, and the admissibility diagnostics are the instrument for detecting strain.

**Admissibility pressure is high.** In U, 66.6% of unconstrained Gaussian probe draws would violate the imposed region, and 1.6% of proposed latent paths are rejected by whole-path positivity; R1 is comparable. The probe share does not enter the posterior — it is a measurement of how much the constraints are doing — but a share this high is exactly the signal the diagnostic was built to raise, and every moving-reference model is flagged for review on it.

**$\gamma$  and  $\zeta_0$  are not identified by this cell.**  $\gamma$ 's posterior tracks its prior, and the implied elasticity has  $P(\zeta_0 > 1) = 0.48$ . Claims that rest on either — the weak Third law, or a markup read off the estimated elasticity — are not supported by this data cell, whatever the restricted models' point estimates look like.

| Model | Probe nonadmissible % | State reject % | Stored kappa $\leq$ 0 % | Stored theta $\leq$ 0 % | Local-range flag |
|-------|-----------------------|----------------|-------------------------|-------------------------|------------------|
| F0    | —                     | —              | 0.0000                  | 0.0000                  | none             |
| U     | 66.5625               | 1.6416         | 0.0000                  | 0.0000                  | review           |
| R1    | 62.9569               | 1.4316         | 0.0000                  | 0.0000                  | review           |
| R2    | 78.3520               | 1.1239         | 0.0000                  | 0.0000                  | review           |
| R3    | 55.0403               | 0.6333         | 0.0000                  | 0.0000                  | review           |

Table 4: Local-model admissibility diagnostics. The probe column counts unconstrained conditional draws that would violate the admissible region; it is a diagnostic and does not enter the posterior.

## 9 Robustness

Two robustness dimensions are settled in this revision and two are not.

**Settled: observation design.** Section 7 estimates the same cell under both inflation transformations for the two models entitled to it.

**Settled: restriction depth.** The hierarchy is itself a robustness exercise. R1 against U isolates the cross-equation identity; R2 against R1 isolates the weak Third law; R3 against R1 isolates constant pass-through. All four models are estimated on identical data, priors, and sample, and the builder refuses to tabulate them together if any of those differ.

**Not settled: prior sensitivity.** The historical report varies the prior set (**tight**, **weak**) across every cell. The restriction namespace has only **baseline** runs. Given that  $\gamma$ 's posterior is prior-dominated and R1's  $\kappa_N$  prior is induced rather than stated, a prior sweep is the single most valuable missing robustness check here, not an optional extra.

**Not settled: sampled  $\zeta_0$ .**  $\zeta_0$  is fixed at 6.00 throughout. Section 5 shows the data cannot pin it, so a sampled- $\zeta_0$  specification would need either a different data cell or an informative external prior.

## 10 Conclusion

On this data cell, the local Rotemberg–HSA cross-equation restriction is neither rejected nor confirmed — and the reason is identification, not tension. The unrestricted model’s posterior for  $\kappa_N$  is centred near zero with a wide two-sided interval that comfortably contains the restricted value, so the restriction survives; but the same width means the data would have tolerated almost any value, so surviving is weak evidence. What the restriction does supply is identification. It converts a near-uninformative  $\kappa_N$  into one that is positive and bounded away from zero, at essentially no cost in the fit of the inflation equation and with one fewer free coefficient. The Second law is maintained rather than tested by that route; the weak Third law is maintained rather than tested outright, since  $\gamma$  is prior-dominated. The fixed-reference benchmark F0 is the quiet result. It fits as well as anything else here with the fewest coefficients and no moving reference at all. Any claim that the moving-reference machinery earns its keep needs the marginal likelihood that this revision does not yet compute.

## 11 Status and open items

1. **Marginal likelihood for the theory namespace.** Chib's method is implemented for the historical models only. Without it there is no formal test of the nested restrictions, and Section 6 must be read as a conditional in-sample diagnostic. This is the largest gap.
2. **Prior robustness sweep.** `tight` and `weak` runs for all five models; required to separate the restriction's effect from its induced prior on  $\kappa_N$ .
3. **U is not literally unrestricted.** With path positivity on by default, U maintains  $\kappa_t > 0$  and  $\theta_t > 0$  at every date, so its model definition records `second_law: false` while the domain condition is doing part of the same work. The  $U \rightarrow R1$  comparison isolates the cross-equation identity, not the Second law.
4. **A four-quarter aggregation likelihood.** Until four quarterly structural equations can be aggregated without changing coefficient meaning, R1–R3 remain Q/Q-only.
5. **Admissibility pressure.** Every moving-reference model is flagged for local-range review. Either the sample must be split into ranges over which the linearisation is defensible, or the approximation must be replaced.
6. **R4.** The fixed-reference identities  $\kappa_0 = (\zeta_0 - 1)/\chi$  and  $\theta_0 = (10/\chi)(1 - \rho_0)/\rho_0$  remain deliberately un-imposed.

## A Appendix: convergence diagnostics

Across all five models the worst scalar bulk ESS is 597 at a maximum  $\hat{R}$  of 1.010, and the worst latent-path bulk ESS is 843 at 1.006. The binding parameter is  $\rho_2$  in the moving-reference models — the second AR(2) coefficient of the competition cycle — which is why those chains are run two orders of magnitude longer than F0’s.

| Model | Scalar min ESS | Scalar max R-hat | Worst scalar | Path min ESS | Path max R-hat | Status    |
|-------|----------------|------------------|--------------|--------------|----------------|-----------|
| F0    | 597            | 1.000            | sigma_u      | 2511         | 1.002          | converged |
| U     | 1152           | 1.010            | rho_2        | 1273         | 1.006          | converged |
| R1    | 1168           | 1.000            | rho_2        | 1332         | 1.002          | converged |
| R2    | 1056           | 1.000            | rho_2        | 1150         | 1.003          | converged |
| R3    | 610            | 1.000            | sigma_eps    | 843          | 1.005          | converged |



## B Appendix: every run behind this report

Estimation revision: 2026-08-moving-reference-hsa-v1; estimation code: 204a2f319b88; sample: 1982Q1–2012Q4; report builder code: 9f7c895a8026; builder dirty: yes.

| Model | Run ID          | Inflation | T   | Status    |
|-------|-----------------|-----------|-----|-----------|
| F0    | 20260810_175459 | qoq       | 124 | converged |
| R1    | 20260810_175459 | qoq       | 124 | converged |
| R2    | 20260810_175459 | qoq       | 124 | converged |
| R3    | 20260810_175657 | qoq       | 124 | converged |
| U     | 20260810_175459 | qoq       | 124 | converged |
| F0    | 20260810_180342 | yoy_4q    | 124 | converged |
| U     | 20260810_180540 | yoy_4q    | 124 | converged |

The run manifest, together with each run’s own `metadata.json`, records code revision, model definition and hierarchy, exact restrictions, maintained laws, admissibility conditions, inflation and expectation transformations, activity and competition proxies,  $N_0$ , the  $\zeta_0/\mu_0$  treatment, sample, sampler, chains, seeds, and convergence. The build fails if a comparison would mix incompatible revisions, samples, transformations, observation designs, proxies, priors, or model definitions. Neither the historical report nor `results/runs` is read by this builder.

## C Appendix: prior against posterior

The panels below are the evidence for two claims made in the body: that  $\gamma$  is prior-dominated in every model that leaves it free, and that  $\kappa_N$ 's posterior in R1–R3 is materially narrower than U's because the restriction supplies information the data do not.

Prior against posterior, every model and restriction-relevant coefficient

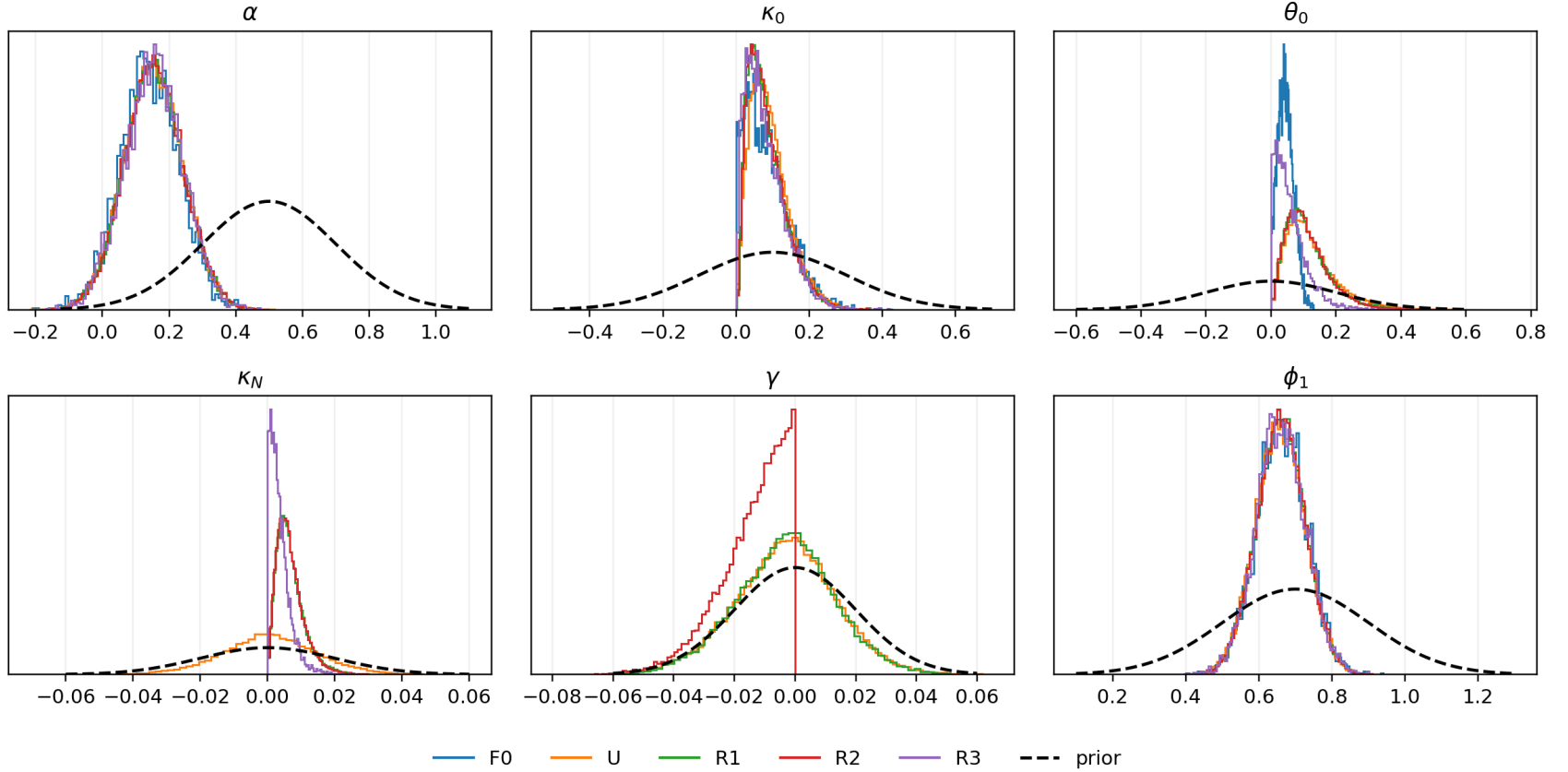


Figure 7: Prior against posterior, every model and restriction-relevant coefficient. In panels where a model has no such coefficient it is simply absent; colours are fixed per model across all panels.